

VMC-3000 Series | MID 128 | Subsystem Code: SPN

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1. Introduction

This manual covers the complete spindle drive system of the VMC-3000 vertical machining centre. The spindle system (subsystem code SPN, Module Identification MID 128) is responsible for rotating the cutting tool at programmed speeds from 100 to 8,000 RPM and transmitting torque to the workpiece material. This document provides component descriptions, parameter specifications, wiring references, error code definitions, maintenance procedures, and spare parts lists.

WARNING: Spindle maintenance must only be performed by trained service engineers. Incorrect bearing installation voids warranty and risks catastrophic spindle failure. Always perform LOTO before accessing spindle internals.

2. System Description

2.1 Spindle Motor

The VMC-3000 spindle motor is a 3-phase AC induction motor with integrated thermal protection (PTC thermistor embedded in stator winding). The motor is directly coupled to the spindle shaft via a rigid coupling — no belt or gearbox. Motor nameplate data:

Parameter	Value
Manufacturer	EquipmentIQ Drive Systems
Model	SPN-18-8000-ISO40
Rated Power (S1)	18.5 kW
Peak Power (S6-25%)	22.0 kW
Rated Speed	4,500 RPM
Maximum Speed	8,000 RPM
Rated Torque	39 Nm
Peak Torque	145 Nm
Rated Current	38 A
Motor Frame	IEC 200L
Cooling Method	IC416 (forced air, external fan)
Insulation Class	F (155 °C)
Encoder Type	2,048 PPR incremental + absolute zero mark
Bearing Set (front)	2× angular contact 7014 CDBP4
Bearing Set (rear)	1× cylindrical roller NU1014
Grease Type	Kluber Isoflex NBU 15
Grease Repack Interval	4,000 h or 2 years (whichever earlier)

Table 2-1: Spindle Motor Nameplate Data

2.2 Spindle Drive Inverter

The spindle motor is controlled by a regenerative AC drive inverter (model SPN-DRV-22KW) mounted in the electrical cabinet. The drive receives speed commands from the CNC controller via high-speed serial link (FSSB) and regulates motor current to achieve commanded speed with closed-loop feedback from the spindle encoder. The drive monitors output current (P002), motor temperature via PTC input (P003), and DC bus voltage (P090).

2.3 Condition Monitoring — Bosch CISS Accelerometer

A Bosch CISS tri-axial MEMS accelerometer is permanently mounted at the spindle nose housing (refer to Figure 2-1 in installation drawing EIQ-DWG-SPN-003). Accelerometer specifications:

Parameter	Specification
Sensor Model	Bosch CISS (Connected Industrial Sensor Solution)
Measurement Range	±8 g (each axis)
Sampling Rate	2,000 Hz (2 kHz) per axis
Axes Monitored	X (lateral), Y (longitudinal), Z (axial)
Output Format	16-bit signed integer, units: raw ADC counts
Scaling	1 count = 1 mg (approximately)
Interface	Bluetooth 4.2 / USB (commissioning mode)
Operating Temperature	-20 °C to +70 °C
IP Rating	IP67
Mounting Torque	12 Nm (M6 stainless bolts)

Table 2-2: Bosch CISS Accelerometer Specifications

3. Monitored Parameters

The following parameters are monitored by the SPN subsystem. Parameter IDs (PID) follow SAE J2012 convention. All values are available on the OPC-UA server at node ns=2;s=SPN.{param_id}.

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P001	1	Spindle Speed	RPM	100 – 8000	0 – 8500
P002	2	Spindle Load	%	0 – 85	0 – 105
P003	3	Spindle Motor Temperature	°C	20 – 75	0 – 95
P004	4	Spindle Bearing Vibration	mm/s	0 – 4.5	0 – 11.2
P005	5	Spindle Torque	Nm	0 – 120	0 – 145
P006	6	Spindle Orientation Error	deg	-0.05 – 0.05	-0.5 – 0.5
P060	60	X-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2
P061	61	Y-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2
P062	62	Z-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P063	63	Vibration Crest Factor	-	1.0 – 4.5	0 – 12.0
P064	64	Kurtosis Index	-	2.5 – 5.0	0 – 25.0

Table 3-1: SPN/VIB Monitored Parameters — VMC-3000

4. Error Codes — SPN Subsystem

The following error codes are defined for the Spindle Drive System (MID 128). Codes follow the format SPN-{SEVERITY}-{NUMBER}. Severity levels 1–8 correspond to CRITICAL through ADVISORY. All codes are indexed in the EquipmentIQ error database and RAG system.

Error Code	Severity	Title	Primary Param	Action
SPN-CR-001	CRITICAL	Spindle Bearing Catastrophic Failure	—	Immediate machine stop. Do not restart with
SPN-CR-002	CRITICAL	Spindle Drive Overcurrent — Hard Fault	—	Immediate machine stop. Do not restart with
SPN-CR-003	CRITICAL	Spindle Orientation Failure — Tool Change Blocked	—	Immediate machine stop. Do not restart with
SPN-MJ-001	MAJOR	Spindle Motor Overtemperature	—	Stop current operation. Machine must be in
SPN-MJ-002	MAJOR	Spindle Bearing Vibration — Major Threshold Exceeded	—	Stop current operation. Machine must be in
SPN-MJ-003	MAJOR	Spindle Load Anomaly — Possible Tool Breakage	—	Stop current operation. Machine must be in
SPN-MJ-004	MAJOR	Regenerative Chatter — Stability Limit Exceeded	—	Stop current operation. Machine must be in
SPN-MJ-005	MAJOR	Spindle Drive Thermal Overcurrent	—	Stop current operation. Machine must be in
SPN-SR-001	SERIOUS	Spindle Load Gradually Increasing — Tool Wear Trend	—	Complete current cycle then stop. Schedule
SPN-SR-002	SERIOUS	Spindle Speed Deviation — Encoder Signal Degraded	—	Complete current cycle then stop. Schedule
SPN-SR-003	SERIOUS	Bearing Kurtosis Threshold — Defect Initiation Detected	—	Complete current cycle then stop. Schedule
SPN-MD-001	MODERATE	Spindle Load — Elevated Baseline	—	Log fault. Schedule maintenance within 24
SPN-MD-002	MODERATE	Feed Override Active — Non-Standard Cutting Condition	—	Log fault. Schedule maintenance within 24
SPN-MN-001	MINOR	Spindle Load — Upper Advisory Threshold	—	Log fault. Monitor parameters. Schedule in
SPN-MN-002	MINOR	Spindle Motor Temperature — Elevated	—	Log fault. Monitor parameters. Schedule in
SPN-MN-003	MINOR	Spindle Torque — Elevated During Entry Move	—	Log fault. Monitor parameters. Schedule in
SPN-MN-004	MINOR	Vibration RMS — Gradual Increase Detected	—	Log fault. Monitor parameters. Schedule in
SPN-WN-001	WARNING	Spindle Load — Tool Life 80% Consumed	—	Log event. No immediate action required. F
SPN-WN-002	WARNING	Spindle Temperature — Warm-Up Reminder	—	Log event. No immediate action required. F
SPN-WN-003	WARNING	Tool Life Counter — Approaching Limit	—	Log event. No immediate action required. F
SPN-WN-004	WARNING	Spindle Load Variance — Increased Part-to-Part Variation	—	Log event. No immediate action required. F
SPN-NC-001	NOTICE	Spindle Speed Program Limit Active	—	Informational. Parameter approaching limit
SPN-NC-002	NOTICE	Spindle Warm-Up Complete	—	Informational. Parameter approaching limit

Error Code	Severity	Title	Primary Param	Action
SPN-AD-001	ADVISORY	Spindle Running at Reduced Speed — Override Active	—	Performance advisory only. No maintenance required.
SPN-AD-002	ADVISORY	Spindle Idle — No Load Detected	—	Performance advisory only. No maintenance required.

Table 4-1: Spindle Drive System Error Codes (All Severity Levels)

4.1 Critical & Major Error Code Details

SPN-CR-001		Spindle Bearing Catastrophic Failure	CRITICAL
Description	Spindle bearing has reached catastrophic failure threshold. Vibration RMS on spindle exceeds 11.2 mm/s with kurtosis index >20. Continued operation risks spindle shaft damage and workpiece ejection.		
Probable Cause	Bearing race spalling, loss of lubrication, contamination ingress, or end-of-life fatigue. May follow undetected progression through WN-060 and MN-061 warnings.		
Diagnostic Steps	- Stop spindle immediately and lock out/tag out - Measure bearing vibration with handheld analyser at spindle nose - Inspect spindle cooling lines for blockage - Check oil mist lubrication nozzle P040 - Review vibration trend log for P004 over prior 72 hours		
Remedy	Replace spindle bearing set. Re-grease per OEM specification. Run acceptance test program SPD-TEST-001 before production restart.		
Required Action	Immediate machine stop. Do not restart without authorised technician inspection.		
Related Codes	SPN-MJ-002, VIB-WN-060, VIB-MN-061		

SPN-CR-002		Spindle Drive Overcurrent — Hard Fault	CRITICAL
Description	Spindle drive current exceeded 145% of rated value for more than 200ms. Drive has tripped on hardware overcurrent protection. P002 read {val} % (limit 105%).		
Probable Cause	Spindle motor short circuit, drive IGBT failure, or mechanical seizure of spindle bearings. Distinguish from thermal overcurrent SPN-MJ-005.		
Diagnostic Steps	- Do not reset drive until root cause identified - Check spindle motor insulation resistance (megger test) - Inspect drive power stage for burn marks - Verify no mechanical obstruction in spindle bore - Check P005 torque log for sudden spike event		
Remedy	Replace faulty drive module or motor depending on fault localisation. Verify DC bus voltage P090 is within spec before power-on.		
Required Action	Immediate machine stop. Do not restart without authorised technician inspection.		
Related Codes	SPN-MJ-005, ELC-CR-001, SPN-CR-001		

SPN-CR-003		Spindle Orientation Failure — Tool Change Blocked	CRITICAL
Description	Spindle failed to reach orientation position within timeout (5 s). Orientation error P006 = {val} deg (limit ± 0.5 deg). Tool change cycle aborted to prevent ATC arm collision.		
Probable Cause	Encoder signal loss, orientation proximity switch failure, or mechanical brake malfunction.		
Diagnostic Steps	- Check spindle encoder cable continuity - Verify orientation switch target disk alignment - Inspect pneumatic brake actuation pressure - Test orientation cycle manually in JOG mode - Review parameter P082 for last NC block executed		
Remedy	Replace encoder or repair signal cable. Re-teach orientation position parameter. Re-run tool change acceptance cycle TCS-TEST-002.		
Required Action	Immediate machine stop. Do not restart without authorised technician inspection.		
Related Codes	TCS-CR-001, SPN-MJ-003, CNC-MD-011		

SPN-MJ-001		Spindle Motor Overtemperature	MAJOR
Description	Spindle motor winding temperature P003 = {val} °C exceeds major threshold (85 °C). Thermal model predicts critical threshold (95 °C) in <15 minutes at current load.		

Probable Cause	Cooling fan failure, blocked air filter, excessive duty cycle, or degraded motor winding insulation.
Diagnostic Steps	• Check spindle cooling fan rotation • Inspect and clean air filter on motor housing • Verify coolant-through-spindle flow if equipped • Reduce spindle load P002 by lowering feed rate • Check ambient temperature P070
Remedy	Replace cooling fan motor. Clean filters. If repeated, check motor insulation class and compare to duty cycle.
Required Action	Stop current operation. Machine must be inspected before resuming production.
Related Codes	SPN-CR-002, THM-WN-070, SPN-MN-006

SPN-MJ-002	Spindle Bearing Vibration — Major Threshold Exceeded	MAJOR
Description	Spindle bearing vibration P004 = {val} mm/s exceeds ISO 10816-3 Zone C limit (7.1 mm/s). Kurtosis index P064 = {val2} indicates bearing defect frequency component present.	
Probable Cause	Bearing inner or outer race defect, ball spall, or loss of preload. Consistent with tool_wear or spindle_bearing_fault events in operational data.	
Diagnostic Steps	• Capture vibration spectrum via on-machine sensor • Compare dominant frequency to bearing defect frequencies (BPFO, BPFI, BSF) • Check spindle thermal growth compensation active • Inspect tool holder for imbalance • Review trend for P004 over prior 30 days	
Remedy	Schedule spindle bearing replacement within 8 hours. Reduce spindle speed by 30% until replacement. Do not run over 6000 RPM.	
Required Action	Stop current operation. Machine must be inspected before resuming production.	
Related Codes	SPN-CR-001, VIB-MN-061, SPN-SR-003	

SPN-MJ-003	Spindle Load Anomaly — Possible Tool Breakage	MAJOR
Description	Spindle load P002 jumped from baseline by >40% within one NC block. P005 torque = {val} Nm. Pattern consistent with sudden tool breakage during milling operation.	
Probable Cause	Tool fracture due to worn cutting edges, excessive depth of cut, incorrect tool material, or coolant failure at cutting zone.	
Diagnostic Steps	• Halt program and inspect tool in spindle • Check tool length with probe if available • Review P033 coolant level and P030 flow rate • Inspect workpiece surface for gouging • Review NC block G-code for programmed depth of cut	
Remedy	Replace broken or worn tool. Verify tool offset in tool table. Run first-off inspection before resuming production.	
Required Action	Stop current operation. Machine must be inspected before resuming production.	
Related Codes	CLS-SR-001, SPN-WN-004, TCS-MD-001	

SPN-MJ-004	Regenerative Chatter — Stability Limit Exceeded	MAJOR
Description	Vibration pattern on X/Y/Z axes indicates regenerative chatter. Dominant frequency {val} Hz matches spindle-speed-to-tooth-passing ratio. Crest factor P063 = {val2}.	
Probable Cause	Operating at unstable depth-of-cut for current spindle speed and tool geometry. May worsen with tool wear.	
Diagnostic Steps	• Record dominant chatter frequency from vibration log • Apply stability lobe diagram for tool/fixture combination • Check workpiece clamping force • Inspect tool holder taper contact for fretting • Review programmed RPM P001 vs stability lobe chart	

Remedy	Adjust spindle speed by $\pm 10\%$ to shift operating point to stable lobe. Reduce axial depth of cut. Consider high-damping tool holder.
Required Action	Stop current operation. Machine must be inspected before resuming production.
Related Codes	VIB-WN-060, SPN-MJ-002, AXS-MN-002

5. Wiring Reference

The following table defines the key electrical connections for the SPN subsystem. Full wiring diagrams are in drawing set EIQ-DWG-SPN-001 to EIQ-DWG-SPN-008. Wire gauge is AWG/metric equivalent.

Table 5-1: Spindle Drive System — Wiring Connections

Signal Name	From (Module:Pin)	To (Module:Pin)	Wire Gauge	Color Code	Notes
Spindle Motor U-phase	SPN-DRV-22KW:U	MTR-SPN:U1	6 mm2	Brown	Screened power cable
Spindle Motor V-phase	SPN-DRV-22KW:V	MTR-SPN:V1	6 mm2	Black	Screened power cable
Spindle Motor W-phase	SPN-DRV-22KW:W	MTR-SPN:W1	6 mm2	Grey	Screened power cable
Motor PE (ground)	SPN-DRV-22KW:PE	MTR-SPN:PE	6 mm2	Green/Yellow	
Encoder A+	MTR-SPN:ENC-A+	CNC-500:SPN-ENC-A+	0.5 mm2	White	Twisted pair, shielded
Encoder A-	MTR-SPN:ENC-A-	CNC-500:SPN-ENC-A-	0.5 mm2	White/Black	Twisted pair, shielded
Encoder B+	MTR-SPN:ENC-B+	CNC-500:SPN-ENC-B+	0.5 mm2	Green	
Encoder B-	MTR-SPN:ENC-B-	CNC-500:SPN-ENC-B-	0.5 mm2	Green/Black	
Encoder Z (zero mark)	MTR-SPN:ENC-Z	CNC-500:SPN-ENC-Z	0.5 mm2	Blue	
Motor PTC (thermal)	MTR-SPN:PTC+	SPN-DRV-22KW:PTC-IN	0.75 mm2	Red	PTC thermistor, 1330 ohm at 155 C
CISS Sensor Power	CISS-SPN:VCC	EIQ-EDGE:SENS-VCC	0.5 mm2	Red	5 V DC
CISS Sensor GND	CISS-SPN:GND	EIQ-EDGE:SENS-GND	0.5 mm2	Black	
Orientation Switch	SPN-ORIENT-SW:COM	PLC-I/O:X0.4	0.75 mm2	Orange	NPN, normally open
Spindle Brake (24V)	PLC-Q:Y0.3	SPN-BRAKE-S0L:A1	0.75 mm2	Violet	24 V DC coil, 15 W

6. Preventive Maintenance Schedule

Interval	Task	Reference	Duration
Daily	Check spindle warm-up program executed (P003 >35 °C before full-speed cuts)	SPN-WN-002	5 min
Weekly	Inspect spindle taper bore for chips or damage; clean with IPA	Visual	10 min
Monthly	Check vibration trend P004 vs baseline; log result	SPN-SR-003	15 min
500 h	Verify spindle encoder signal quality on oscilloscope	SPN-SR-002	30 min
1,000 h	Clean spindle cooling air filter; check fan rotation	SPN-MJ-001	20 min
2,000 h	Full spindle vibration spectrum analysis (FFT)	DOC-EIQ-005	60 min
4,000 h	Spindle bearing grease repack (Kluber Isoflex NBU 15)	EIQ-PROC-SPN-001	4 h

Interval	Task	Reference	Duration
On alarm	Review P004 trend; escalate per SPN-MJ-002 or SPN-CR-001	Error code	—

Table 6-1: Spindle Preventive Maintenance Schedule

7. Spare Parts List

Part Number	Description	Qty	Unit	Lead Time	Safety Stock
EIQ-SPN-BRG-7014	Angular contact bearing 7014 CDBP4 (front, pair)	1	Pair	2 weeks	2
EIQ-SPN-BRG-NU1014	Cylindrical roller bearing NU1014 (rear)	1	Each	1 week	1
EIQ-SPN-ENC-2048	Spindle encoder, 2048 PPR, incremental + Z	1	Each	3 weeks	1
EIQ-SPN-FAN-200L	Spindle motor cooling fan (IC416)	1	Each	1 week	2
EIQ-SPN-FILTER-001	Spindle motor air filter mat (pack of 5)	1	Pack	1 week	2
EIQ-SPN-GREASE-NBU15	Klüber Isoflex NBU 15 grease, 50 g cartridge	1	Each	3 days	5
EIQ-SPN-BRAKE-SOL	Spindle orientation brake solenoid, 24VDC 15W	1	Each	2 weeks	1
EIQ-CISS-SENSOR-001	Bosch CISS tri-axial accelerometer (replacement)	1	Each	4 weeks	1
EIQ-SPN-DRV-22KW	Spindle drive inverter 22 kW (full replacement)	1	Each	6 weeks	1
EIQ-SPN-CABLE-POWER	Spindle motor power cable, 6mm ² screened, 5m	1	Each	1 week	2

Table 7-1: Spindle System Spare Parts